Automated Annotation of Complex Natural Products Using A Modular Fragmentation-based Structure Assembly (MFSA) Strategy

User Manual of CNPs-MFSA

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Introduction

Complex natural products (CNPs) Structure Annotation Tool is a user-friendly application for processing and analyzing LC-MS spectra. It provides functionalities to recognize target natural products, calculate molecular compositions, determine possible adduct ions, identify characteristic neutral losses, extract feature ions, calculate cosine similarity scores between standard and query spectra, and provide possible structure annotation. The tool is designed for both beginners and experts in natural product chemistry, offering an intuitive interface to streamline natural product analysis tasks while maintaining the flexibility for customization.

Installation and Setup

1. Application Requirements:

- Python 3.8 or higher. (Recommend Python 3.11)
- Required Python libraries:

a. Data processing and file handling:

`io, os, re, uuid, base64, zipfile, pandas, numpy, typing, concurrent.futures`

b. Chemical analysis and mass spectrometry data handling:

`sqlite3, pyteomics, matchms, rdkit`

c. Building and managing a web-based dashboard using Dash:

`dash, dash_bootstrap_components, plotly, flask`

2. Installation Steps:

- Download and install Pycharm with Python (Recommend) or other IDE like Anaconda
- Clone the repository:

```
git clone <https://github.com/MiZhang47/CNPs-MFSA.git >
```

- Navigate to the directory: `CNPs-MFSA`

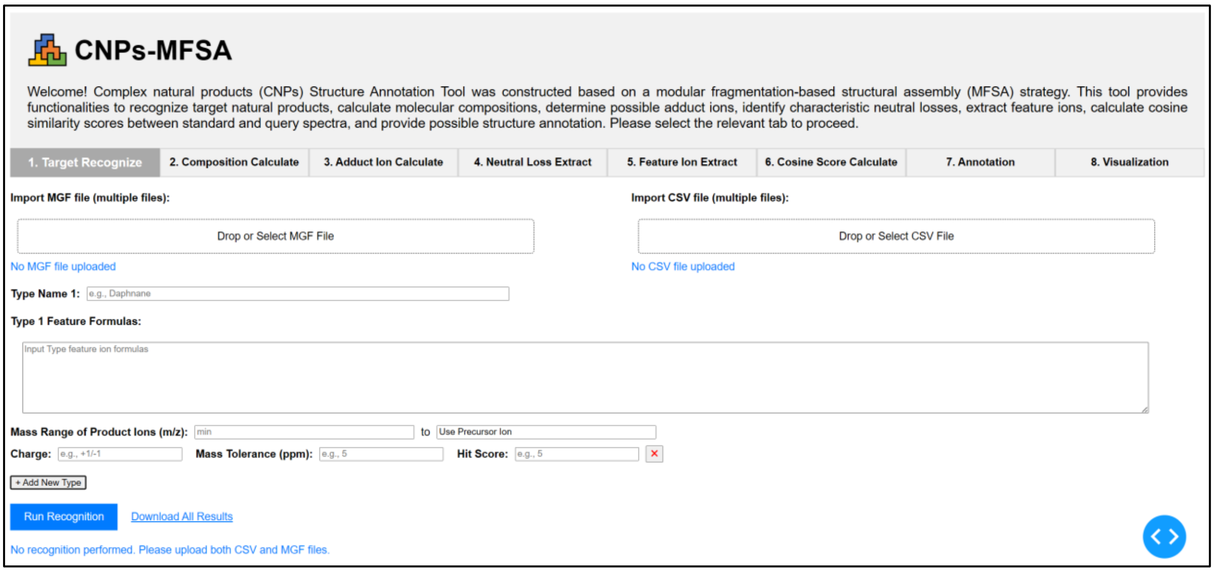
3. Running the Application:

- Start the application by executing: `CNPs-MFSA_Web.py`
- Open a web browser and navigate to: `http://127.0.0.1:8050`

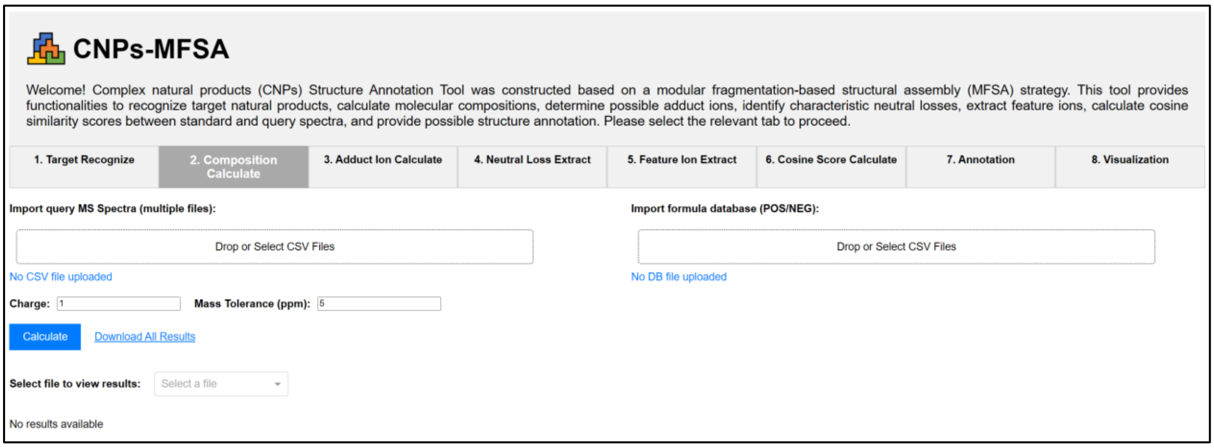
Using the Tool

1. Tabs Overview

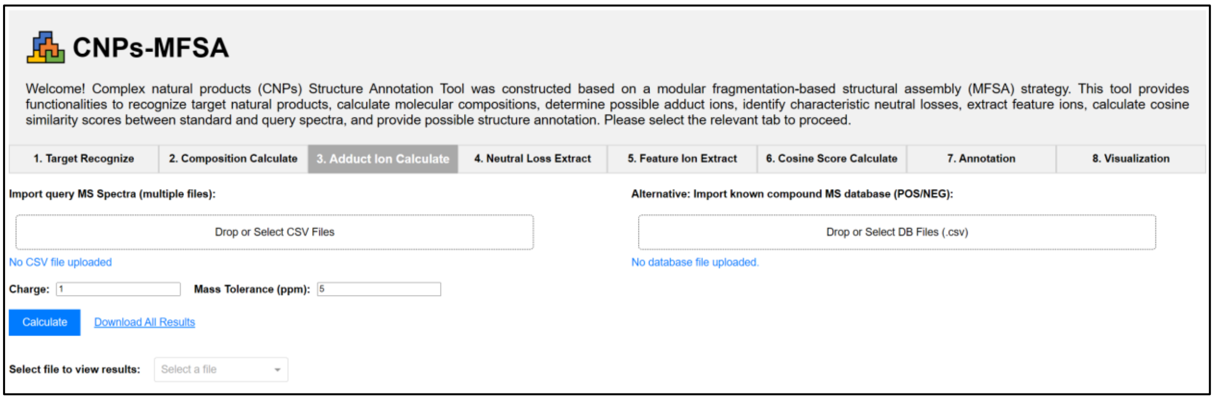
1.1. Target Recognition: Extract specific target precursor ion from uploaded data.



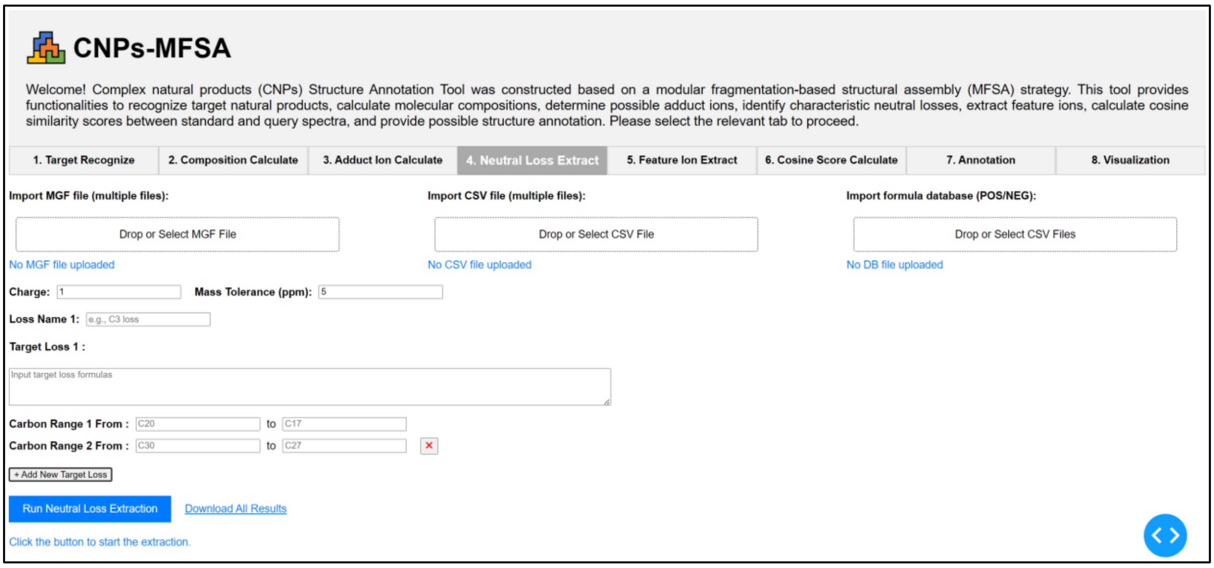
1.2. Composition Calculation: Analyze molecular composition from spectra.



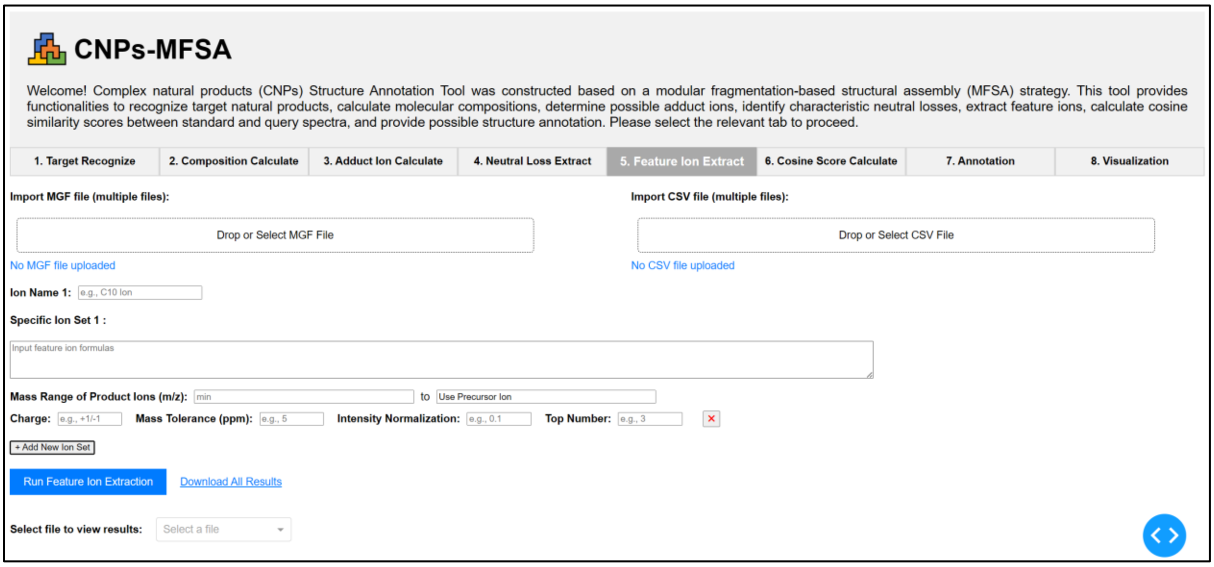
1.3. Adduct Ion Calculation: Identify possible adduct ions in mass spectra.



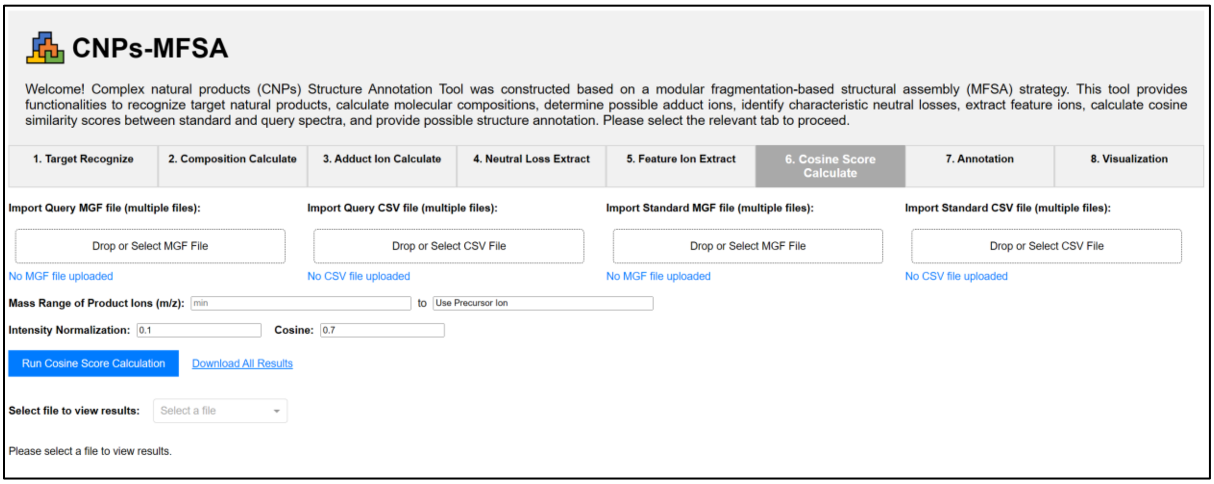
1.4. Neutral Loss Labeling: Identify neutral loss patterns in spectra.



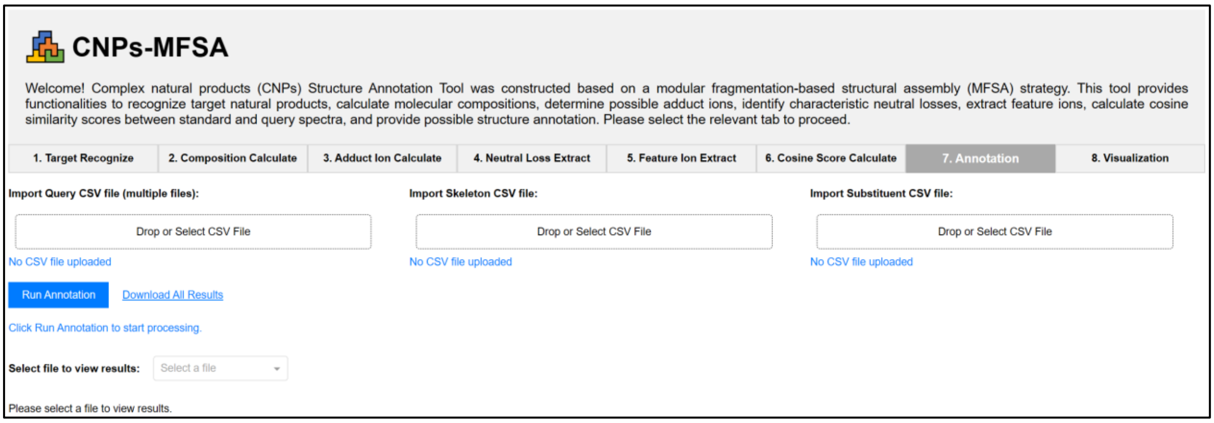
1.5. Feature Ion Labeling: Extract and analyze feature ions.



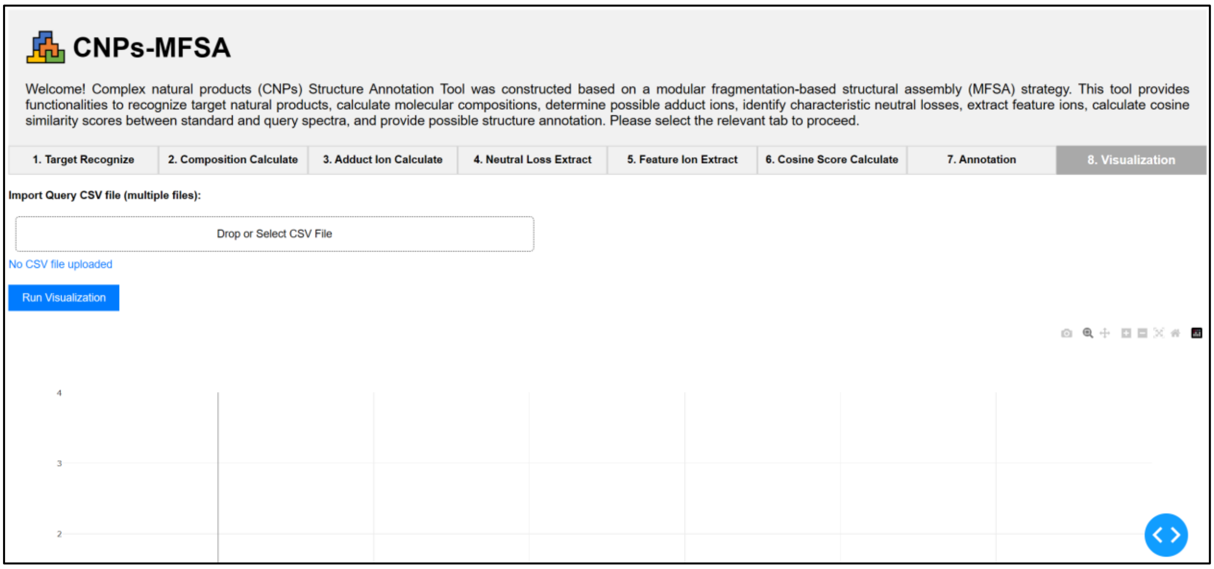
1.6. Cosine Score Calculation: Calculate cosine similarity scores for spectral matching.



1.7. Annotation for daphnane (Alternative)



1.8. Visualization



2. Example Workflow for Daphnane-type Diterpenoids

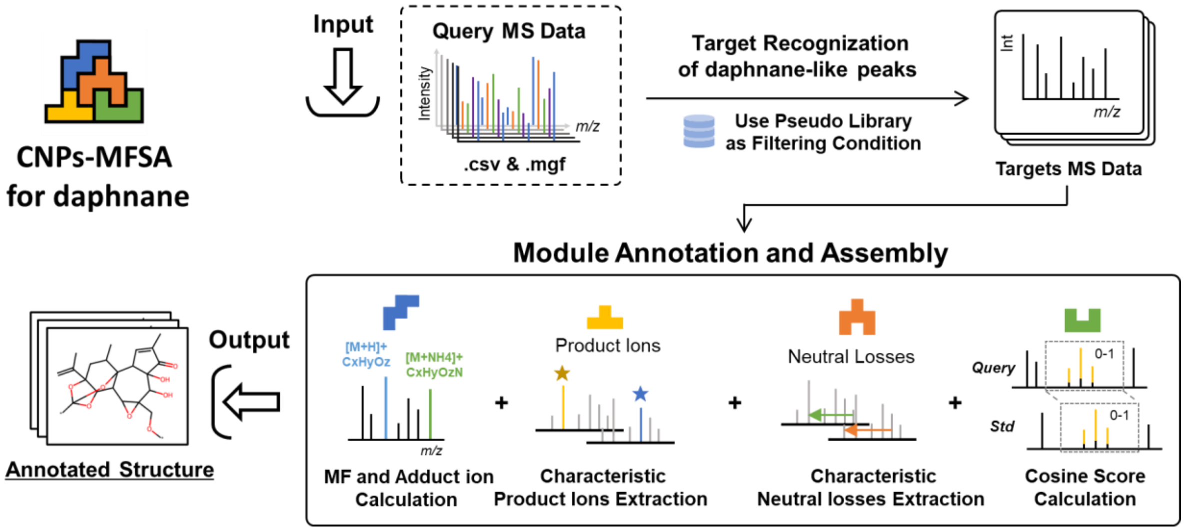


Figure Automatic Annotation Workflow of Thymelaeaceae Daphnanes Using Modular Fragmentation-based Structure Assembly (MFSA) Strategy.

<u>Step 1: Target Recognize</u>

(1) Upload Files: Select MGF and CSV files processed by MZmine

***Note**: In order to speed up the process, we recommended to process raw files as **centroid** data. Before uploading, all **.mgf** files generated by MZmine need to be opened with Notepad, and all **FEATURE_ID** need to be changed to **TITLE** using the search and replace mode.

```
BEGIN IONS                        BEGIN IONS
FEATURE_ID=1          →           TITLE=1
RTINSECONDS=293.536               RTINSECONDS=293.536
CHARGE=1+                         CHARGE=1+
MSLEVEL=2                         MSLEVEL=2
MERGED_STATS=1 / 20               MERGED_STATS=1 / 20
```

(2) Define Parameters: Set up type number, type name, mass range, charge and feature formulas (characteristic product ions).

As for daphnane-type diterpenoids from Thymelaeaceae, the parameters were set as following:

- <u>Type Name 1</u>: **D** [Abbreviation for normal daphnane-type diterpenoids]
- <u>Type 1 Feature Formulas</u>: (The following ion formula were composed of the measured product ion formula of standard normal daphnanes within the mass range and the predicted product ion formula in the pseudo-daphnane library without duplication.)

$C_{20}H_{28}O_4$, $C_{20}H_{28}O_5$, …$C_{18}H_{16}O_2$, $C_{18}H_{16}O_1$, $C_{18}H_{16}$, $C_{17}H_{18}O_1$, $C_{17}H_{16}O_1$, $C_{15}H_{18}O_2$

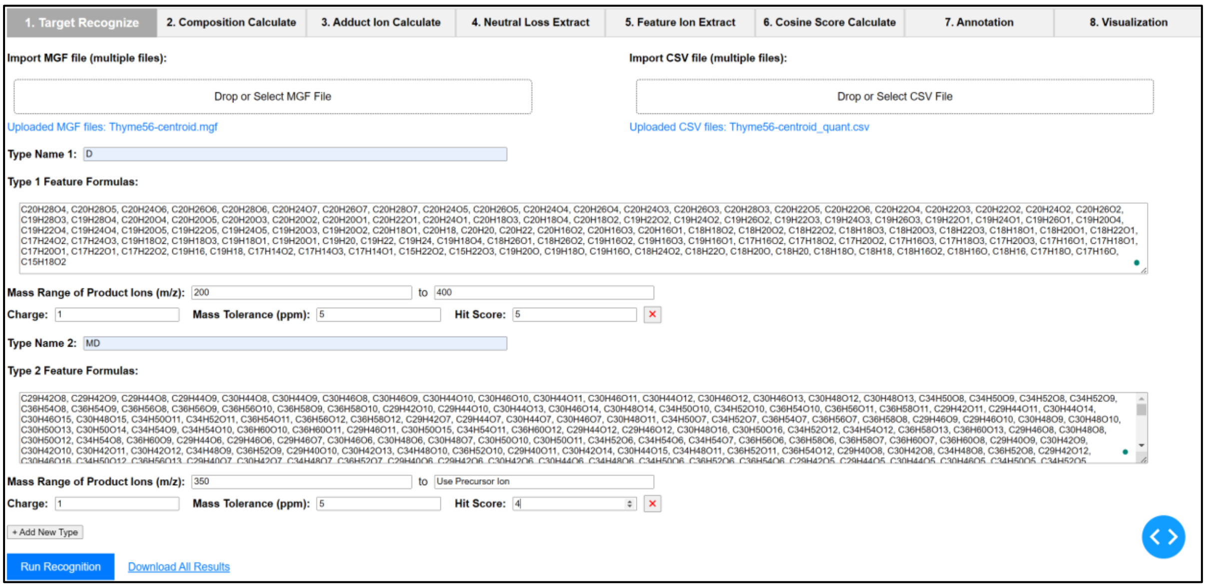
***Note**: For detailed ion information, see the **feature ion and neutral loss.txt** file, which can be find in GitHub repository CNPs-MFSA.

- Mass Range of Product Ions (*m/z*): **200** to **Use Precursor Ion** (Considering that the product ions derived from certain aglycones may be affected by the intensity of the substituent ions, resulting in a decrease in relative intensity, it is possible to normalize the product ions within a specific range to search for specific product ions.)
- Charge: **+1** (positive ion mode)
- Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)
- Hit Score: **5** (The minimum score threshold for normal daphnane-type diterpenoids)
- Type Name 2: **MD** [Abbreviation for macrocyclic daphnane orthoesters]
- Type 2 Feature Formulas: (The following ion formula were composed of the measured product ion formula of standard macrocyclic daphnane orthoesters within the mass range and the predicted product ion formula in the pseudo-daphnane library without duplication.)

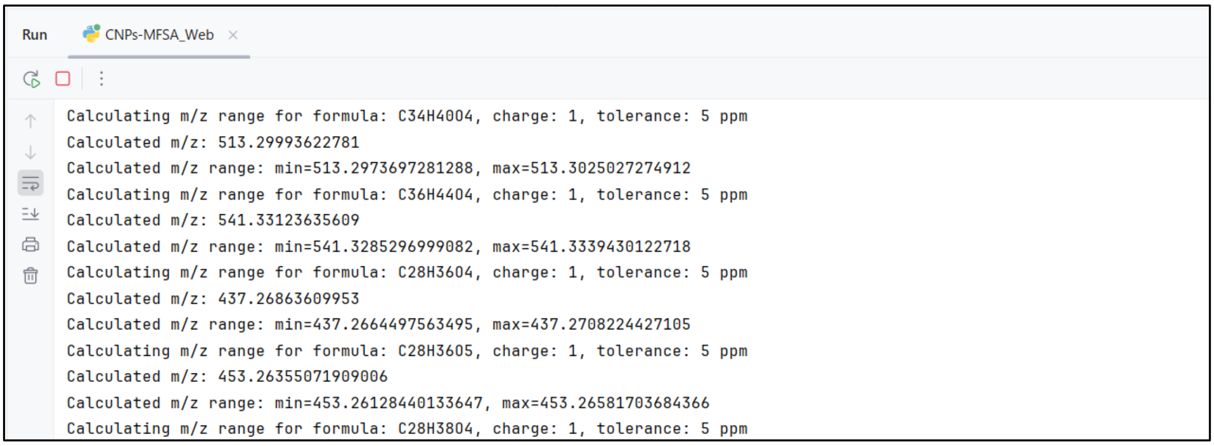
$C_{30}H_{44}O_8$, $C_{30}H_{44}O_9$, …$C_{30}H_{46}O_8$, $C_{28}H_{34}O_9$, $C_{32}H_{38}O_5$, $C_{27}H_{30}O_2$, $C_{32}H_{38}O_2$, $C_{27}H_{32}O_6$

***Note**: For detailed ion information, see the **feature ion and neutral loss.txt** file, which can be find in GitHub repository CNPs-MFSA.

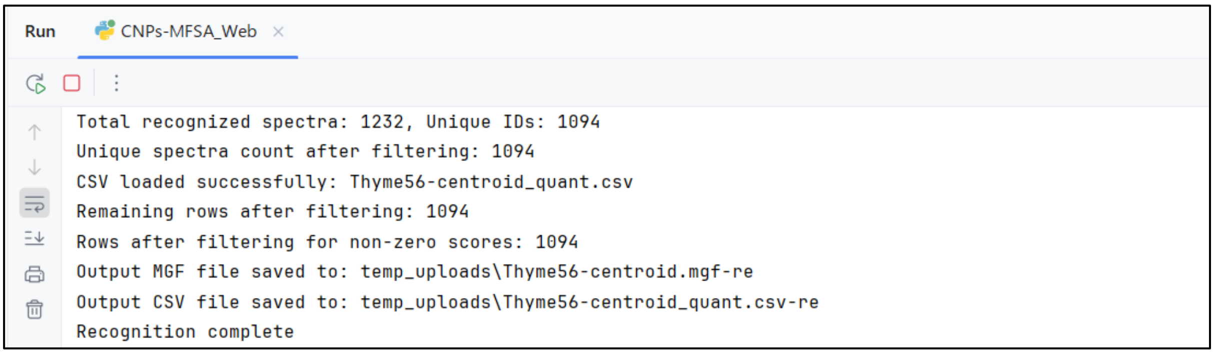
- Mass Range of Product Ions (*m/z*): **350** to **Use Precursor Ion** (same as above)
- Charge: **+1** (positive ion mode)
- Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)
- Hit Score: **4** (The minimum score threshold for normal daphnane-type diterpenoids)



(3) Run the Process: Click "Run Recognition" to execute the analysis, and the python console would be displayed the following information:

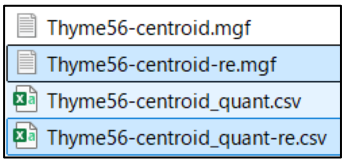


Recognition was complete with the following information:



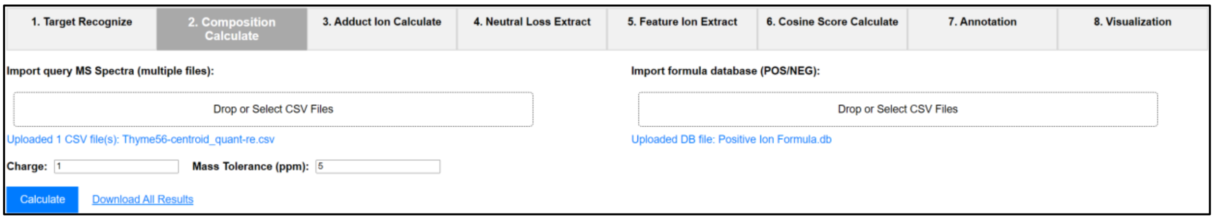
***Note**: On a personal computer, the processing speed of **centroid data** is about **120 precursor ions/minute**, so it takes 100 minutes to process a test data containing about 12,000 precursor ions; while the processing speed of **profile data** is about **60 precursor ions/minute**, it takes 200 minutes to complete the processing of the same test data.

(4) Download Results: download the generated .zip file and the results were shown below with new rows, namely **type score** and **type**. The **xxx-re.csv** was used as the **input file** and switched to the next tab "Composition Calculate". The **xxx-re.mgf** was used as the **input file** of **Step 4**, **5**, and **6**.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	row ID	row m/z	row retention time	D Score	MD Score	type	20221207_	20221221_	20221207_	20221105_	20221105_	20240421_	20221207_	20221207_	20221105_	20240421_	20221105_	20240428_	20240421_	20230410_
2	2	543.2222	4.4936056	16	0	D	4.1E+08	0	0	0	0	0	0	0	0	0	0	0	0	0
3	6	635.321	12.122931	9	0	D	2.79E+08	22649791	0	2799330	58690113	17138181	0	0	2.5E+08	0	2510044	0	0	0
4	9	605.2379	7.003309033	18	0	D	84438393	1189480	0	0	0	0	0	12614013	0	0	12450572	0	0	0
5	11	647.285	10.60498556	11	0	D	1.04E+08	9550283	0	1.8E+08	0	0	0	0	1.24E+09	0	0	0	0	0
6	12	649.3002	10.68500329	12	0	D	1.04E+08	0	0	0	15626314	5556343	0	12159979	6395130	0	53544545	0	0	0
7	14	603.2223	6.764385931	12	0	D	59484555	1937045	0	0	4139901	50246559	2508607	69260852	2.82E+08	0	43993539	0	0	0
8	16	585.2692	8.197227246	15	0	D	57059457	24883537	0	1295131	23654689	0	0	5398647	0	0	4478854	0	0	1294504
9	17	541.207	4.297588633	13	0	D	51765693	0	0	3.93E+08	0	0	0	0	0	0	0	0	0	0
10	21	585.3421	13.57017769	10	0	D	40379331	0	0	0	0	0	0	0	0	0	0	0	0	0
11	25	637.3368	12.87550885	10	0	D	37207731	6938174	0	0	11453971	0	0	18376942	0	0	0	0	0	0
12	31	649.2999	10.17540788	16	0	D	32041569	3050398	0	0	0	0	0	0	0	0	12813287	0	0	0
13	39	591.2585	8.793285861	13	0	D	24872914	8167630	6593211	0	4638095	2072423	0	17605572	18568763	1593591	5.45E+08	0	0	0
14	40	559.2173	2.5230582	22	0	D	24089155	0	0	0	0	0	0	0	0	0	0	0	0	0
15	47	487.2325	4.31315775	11	0	D	19491187	5766567	0	0	0	2126725	0	0	18743957	0	0	0	0	3839010
16	49	653.3318	11.29813031	9	0	D	25719766	8935738	0	0	0	7040123	0	12854141	26020437	0	2510878	0	0	0
17	50	485.2166	2.9796957	5	0	D	12863434	0	0	0	0	0	0	0	0	0	0	0	0	0

Step 2: Composition Calculate



(1) Upload CSV Files: Select **xxx-re.csv** files processed by **Step 1**.

(2) Upload DB Files: Select **Positive Ion Formula.db** files generated by **formula DB generation.py**, which could be found in Github.

***Note**: **formula DB generation.py** generates molecular formulas based on carbon (C), hydrogen (H), and oxygen (O) atoms, as well as Index of Hydrogen Deficiency (IHD) to determine valid formulas, then saves them to an SQLite database. It also simulates the addition of adduct ions, such as $[M+NH_4]^+$ and $[M+HCOO]^-$, to molecules and stores them in separate databases for positive and negative ionization modes. Users could modify `generate_organic_compounds()` to include additional elements or change element number, and adjust IHD range for different constraints. Users also could extend `add_additional_groups()` to include other adduct ions.

```
# Generate a list of organic compounds based on possible combinations of C, H, and O
def generate_organic_compounds():  1 usage
    compounds = []  # List to store the generated compounds
    # Iterate through possible numbers of carbon atoms (from 1 to 60)
    for nC in range(1, 81):
        # Iterate through possible numbers of hydrogen atoms (from 2 to 80, in steps of 2)
        for nH in range( 2, 91, 2):
            # Iterate through possible numbers of oxygen atoms (from 1 to 20)
            for nO in range(1, 26):
                IHD = calculate_IHD(nC, nH, nO)  # Calculate the IHD for the compound
                # Only store compounds with an IHD between 3 and 25 (inclusive)
                if 3 <= IHD <= 25:
                    compound = (nC, nH, nO)
                    compounds.append(compound)
    return compounds

# Simulate the addition of functional groups to a list of compounds
def add_additional_groups(compounds):  1 usage
    # Add an NH3 group to each compound
    nh3_added = [(nC, nH + 3, nO, 1) for nC, nH, nO in compounds]
    # Add a CH2O2 group to each compound
    ch2o2_added = [(nC + 1, nH + 2, nO + 2) for nC, nH, nO in compounds]
    # Add a CH3NH2 group to each compound
    ch5n_added = [(nC + 1, nH + 5, nO, 1) for nC, nH, nO in compounds]
    return nh3_added, ch2o2_added, ch5n_added
```

The **Positive Ion Formula.db** file provided in the input data was designed for the calculation of daphnane-type diterpenes, with **carbon atom** numbers ranging from **1** to **80**, hydrogen atom numbers ranging from **2** to **90**, oxygen atom numbers ranging from **1** to **26**, and Index of Hydrogen Deficiency (IHD) values ranging from **3** to **25**. Furthermore, in addition to **$[M+H]^+$**, **$[M+NH_4]^+$**, and **$[M+CH_3NH_2+H]^+$** considered as adduct ion species for positive ion modes.

***Note**: According to the current parameter settings, the element composition can be correctly calculated for peaks with **row m/z** between **400-1300 Da**, but for peaks outside the mass range, about **10%** of the peaks cannot be calculated. Therefore, the author should conduct manual verification and added the molecular formula manually.

Run CNPs-MFSA_Web

```
Calculated m/z for formula 'C18H2O9': 362.97715810695
m/z range for formula 'C18H2O9': min=362.9753432211595, max=362.9789729927405
Calculated m/z range for formula 'C18H2O9': min=362.9753432211595, max=362.9789729927405
Calculating m/z range for formula: C18H2O10, charge: 1, ppm: 5
Calculated m/z for formula 'C18H2O10': 378.97207272651
m/z range for formula 'C18H2O10': min=378.9701778661464, max=378.97396758687364
Calculated m/z range for formula 'C18H2O10': min=378.9701778661464, max=378.97396758687364
Calculating m/z range for formula: C18H2O11, charge: 1, ppm: 5
Calculated m/z for formula 'C18H2O11': 394.96698734607
m/z range for formula 'C18H2O11': min=394.9650125111333, max=394.96896218100676
Calculated m/z range for formula 'C18H2O11': min=394.9650125111333, max=394.96896218100676
Calculating m/z range for formula: C18H2O12, charge: 1, ppm: 5
Calculated m/z for formula 'C18H2O12': 410.96190196563
```

Calculation was complete with the following information:

Run CNPs-MFSA_Web

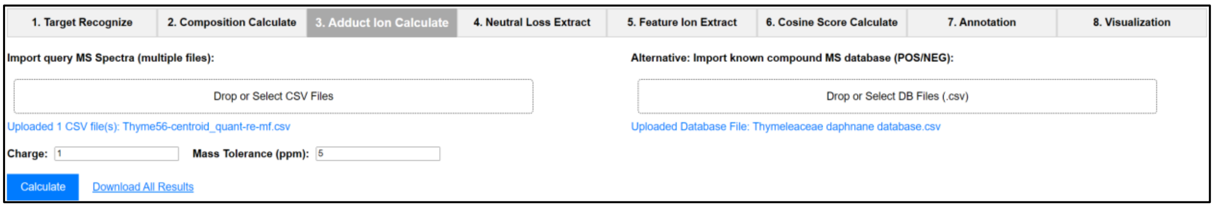
```
Elemental compositions calculated for Thyme56R_quant-re.csv.
Processed CSV saved at: temp\Thyme56R_quant-re.csv
Added Thyme56R_quant-re.csv to ZIP file.
ZIP file created at: temp\composition formula_results.zip
```

(5) Download Results: download the generated **xxx.zip** file and the results were shown below with new row, **<u>elemental composition</u>**. The **xxx-mf.csv** was used as the **input file** and switched to the next tab "Adduct Ion Calculate".

Thyme56-centroid.mgf
Thyme56-centroid-re.mgf
Thyme56-centroid_quant.csv
Thyme56-centroid_quant-re.csv
Thyme56-centroid_quant-re-mf.csv

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	row ID	row m/z	row retention time	D Score	MD Score	type	elemental composition	20221207_	20221221_	20221207_	20221105_	20221105_	20240421_	20221207_	20221207_	20221105_	20240421_	20221105_
2	2	543.2222	4.4936056	16	0	D	C29H34O10	4.1E+08	0	0	0	0	0	0	0	0	0	0
3	6	635.321	12.122931	9	0	D	C37H46O9	2.79E+08	22649791	0	2799330	58690113	17138181	0	0	2.5E+08	0	2510044
4	9	605.2379	7.003309033	18	0	D	C34H36O10	84438393	1189480	0	0	0	0	0	12614013	0	0	12450572
5	11	647.285	10.60498556	11	0	D	C37H42O10	1.04E+08	9550283	0	1.8E+08	0	0	0	0	1.24E+09	0	0
6	12	649.3002	10.68500329	12	0	D	C37H44O10	1.04E+08	0	0	0	15626314	5556343	0	12159979	6395130	0	53544545
7	14	603.2223	6.764385931	12	0	D	C34H34O10	59484555	1937045	0	0	4139901	50246559	2508607	69260852	2.82E+08	0	43993539
8	16	585.2692	8.197227246	15	0	D	C32H40O10	57059457	24883537	0	1295131	23654689	0	0	5398647	0	0	4478854
9	17	541.207	4.297588633	13	0	D	C29H32O10	51765693	0	0	3.93E+08	0	0	0	0	0	0	0
10	21	585.3421	13.57017769	10	0	D	C34H48O8	40379331	0	0	0	0	0	0	0	0	0	0
11	25	637.3368	12.87550885	10	0	D	C37H48O9	37207731	6938174	0	0	11453971	0	0	18376942	0	0	0
12	31	649.2999	10.17540788	16	0	D	C37H44O10	32041569	3050398	0	0	0	0	0	0	0	0	12813287
13	39	591.2585	8.793285861	13	0	D	C34H38O9	24872914	8167630	6593211	0	4638095	2072423	0	17605572	18568763	1593591	5.45E+08
14	40	559.2173	2.5230582	22	0	D	C29H34O11	24089155	0	0	0	0	0	0	0	0	0	0
15	47	487.2325	4.31315775	11	0	D	C27H34O8	19491187	5766567	0	0	0	2126725	0	0	18743957	0	0

Step 3: Adduct Ion Calculate



(1) Upload CSV Files: Select **xxx-mf.csv** files processed by **Step 2**.

(2) **Alternatively upload** known compound MS database (**xxx.csv**) in positive or negative ion modes with the following format for MS1 molecular formula matching:

For example, **Thymeleaceae daphnane database.csv**, which included the **molecular formula (M)** and **composition after possible adducts addition (M+Adducts)** of known daphnanes isolated from Thymeleaceae family (until manuscript submitted) was imported.

***Note**: If users do not require comparison with the established compound molecular formula database, uploading database here was unnecessary. As for the positive ion mode, the composition formula after the addition of the adduct should be entered in the form of *ex*. **$M+NH_3$, rather than $M+NH_3+H$**, due to the charge = +1, and a positive charge would be automatically added during calculation.

	A	B	C	D	E	F
1	**Compound Name**	**Type**	**M**	**M+NH3**	**M+CH3NH2**	**M+⋯**
2	resiniferonol	D	C20H28O6	C20H31O6N1	C21H33O6N	⋯
3	daphneresiniferin A	D	C24H30O8	C24H33O8N1	C25H35O8N	⋯
4	daphnegen B	D	C24H30O10	C24H33O10N1	C25H35O10N	⋯
5	daphnediterp A	D	C25H36O9	C25H39O9N1	C26H41O9N	⋯
6	daphnetoxin (S1)	D	C27H30O8	C27H33O8N1	C28H35O8N	⋯
7	daphne factor F4	D	C27H32O8	C27H35O8N1	C28H37O8N	⋯
8	orthobenzoate 2	D	C27H34O8	C27H37O8N1	C28H39O8N	⋯
9	12-OH-daphnetoxin	D	C27H30O9	C27H33O9N1	C28H35O9N	⋯
10	daphnegiraldigin	D	C27H32O9	C27H35O9N1	C28H37O9N	⋯

(2) Define Parameters: Set up charge and mass tolerance.

- Charge: **+1** (positive ion mode)
- Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)

(3) Adduct Ion Calculation: Click **"Calculate"** to calculate adduct ion species, and the python console would be displayed the following information:

Run CNPs-MFSA_Web

```
Calculated m/z: 615.31637413759, m/z range: (615.3132975557193, 615.3194507194606)
Calculating m/z range for formula: C34H49O10N1, charge: 1, tolerance: 5 ppm
Calculated m/z: 632.3429232386001, m/z range: (632.3397615239838, 632.3460849532163)
Calculating m/z range for formula: C35H50O9, charge: 1, tolerance: 5 ppm
Calculated m/z: 615.35275964631, m/z range: (615.3496828825117, 615.3558364101083)
Calculating m/z range for formula: C35H53O9N1, charge: 1, tolerance: 5 ppm
Calculated m/z: 632.37930874732, m/z range: (632.3761468507763, 632.3824706438637)
Calculating m/z range for formula: C36H40O9, charge: 1, tolerance: 5 ppm
Calculated m/z: 617.27450932561, m/z range: (617.2714229530634, 617.2775956981566)
Calculating m/z range for formula: C36H43O9N1, charge: 1, tolerance: 5 ppm
```

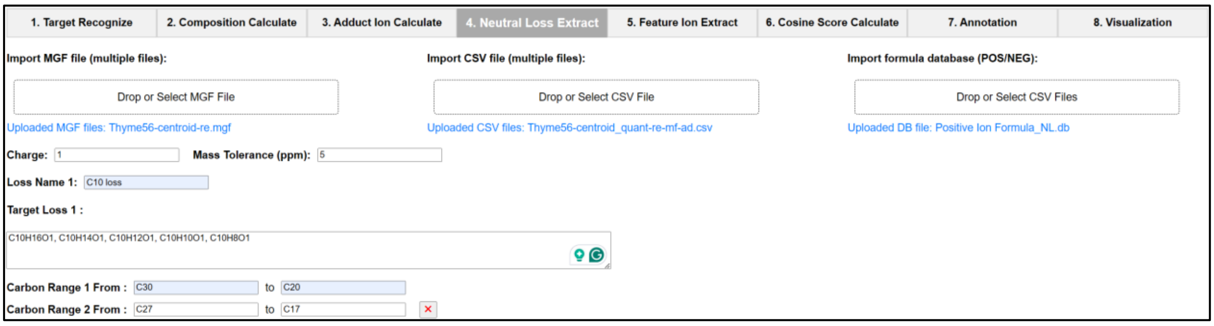
(4) Download Results: download the generated **xxx.zip** file and the results were shown below with new rows, **same peak**, **adduct ion**, **identification**, and **same composition**. The **xxx-ad.csv** was used as the input file and switched to the next tab "Neutral Loss Extract".

- Thyme56-centroid.mgf
- Thyme56-centroid-re.mgf
- Thyme56-centroid_quant.csv
- Thyme56-centroid_quant-re.csv
- Thyme56-centroid_quant-re-mf.csv
- Thyme56-centroid_quant-re-mf-ad.csv

	A	B	C	D	E	F	G	H	I	J	K
1	row ID	row m/z	row retent	D Score	MD Score	type	same peak	elemental composition	adduct ion	identification	same composition
2	1418	680.2911	1.1	8	0	D	none	C33H45O14N1	[M+NH4]+	possibly undescribed	none
3	444	512.2862	1.2	5	0	D	none	C26H41O9N1	[M+NH4]+	possibly known	daphnediterp A
4	9757	682.4177	1.2	0	4	MD	none	C36H59O11N1	[M+NH4]+	possibly undescribed	none
5	3143	682.4152	1.3	0	5	MD	none	C36H59O11N1	[M+NH4]+	possibly undescribed	none
6	10205	469.2223	1.5	6	0	D	10193, 10205	C27H32O7	[M+H-H2O]+	possibly undescribed	none
7	10193	487.2324	1.5	7	0	D	10193, 10205	C27H34O8	[M+H]+	possibly known	orthobenzoate 2
8	1389	680.2908	1.6	13	0	D	none	C33H45O14N1	[M+NH4]+	possibly undescribed	none
9	8407	329.1743	1.7	6	0	D	none	C20H24O4	[M+H]+	possibly undescribed	none
10	7378	374.22	1.7	5	0	D	none	C19H35O6N	[M+NH4]+	possibly undescribed	none
11	10189	451.2111	1.7	6	0	D	none	C27H30O6	[M+H]+	possibly undescribed	none
12	10183	522.2697	1.7	6	0	D	none	C27H39O9N1	[M+NH4]+	possibly known	neogenkwanine I, genkwanine
13	3182	359.1484	1.9	5	0	D	none	C20H22O6	[M+H]+	possibly undescribed	none
14	8456	505.2432	2	11	0	D	none	C27H36O9	[M+H]+	possibly known	neogenkwanine I, genkwanine
15	10405	521.2385	2	7	0	D	none	C27H36O10	[M+H]+	possibly undescribed	none

Step 4: Neutral Loss Labeling

***Notes**: the "Neutral Loss Extract" tab was designed for the accurate calculation of neutral losses within certain carbon ranges. If users need to calculate normal neutral losses among all product ion pairs or between the precursor ion and product ions, the corresponding .py file **normal neutral losses.py** could be found in Github.



(1) Upload MGF Files: Select **xxx-re.mgf** files processed by **Step 1**.

(2) Upload CSV Files: Select **xxx-ad.csv** files processed by **Step 3**.

Upload DB Files: Select **Positive Ion Formula NL.db** files generated by **Formula DB Generation.py** as **Step 2**.

(3) Define Parameters:

a. Set up charge and mass tolerance.

- Charge: +1 (positive ion mode)
- Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)

b. Set up detailed losses and carbon range:

Target Loss 1: **C10 loss**: $C_{10}H_{18}O_2$, $C_{10}H_{16}O_2$, $C_{10}H_{14}O_2$, $C_{10}H_{12}O_2$, $C_{10}H_{10}O_2$

(# Corresponding to the substructure of C10 macrocyclic ring of daphnanes)

Carbon Range 1 From C30 to C20

Carbon Range 2 From C27 to C17

Target Loss 2: **C14 loss**: $C_{14}H_{24}O_2$

(# Corresponding to the substructure of C14 macrocyclic ring of daphnanes)

Carbon Range 1 From C34 to C20

Carbon Range 2 From C34 to C20

Target Loss 3: **C9 loss**: $C_9H_{16}O_2$

(# Corresponding to the substructure of C9 macrocyclic ring of daphnanes)

Carbon Range 1 From C29 to C20

Carbon Range 2 From C29 to C20

Target Loss 4: **C3 loss**: $C_3H_4O_2$

(# Corresponding to the substructure of epoxy moiety of B ring in daphnanes)

Carbon Range 1 From C30 to C27

Carbon Range 2 From C20 to C17

Target Loss 5: **CO2 loss**: $C_1H_0O_2$

(# Corresponding to the substructure of bicyclo[2.2.1]heptane ring A of daphnanes)

Carbon Range 1 From C30 to C29

Carbon Range 2 From C29 to C28

Target Loss 6: **C2 loss**: $C_2H_4O_2$

(# Corresponding to the substructure of acyl substituents of daphnanes)

Carbon Range 1 From C32 to C30

Carbon Range 2 From C22 to C20

Target Loss 7: **C7 loss**: $C_7H_6O_2$

(# Corresponding to the substructure of benzoyl substituents of daphnanes)

Carbon Range 1 From C37 to C30

Carbon Range 2 From C27 to C20

Loss Name 1: C10 loss

Target Loss 1 :

C10H18O2, C10H16O2, C10H14O2, C10H12O2, C10H10O2

Carbon Range 1 From : C30 to C20

Carbon Range 2 From : C30 to C20 ✕

Loss Name 2: C14 loss

Target Loss 2 :

C14H24O2

Carbon Range 1 From : C34 to C20

Carbon Range 2 From : C34 to C20 ✕

Loss Name 3: C9 loss

Target Loss 3 :

C9H16O2

Carbon Range 1 From : C29 to C20

Carbon Range 2 From : C29 to C20 ✕

Loss Name 4: C3 loss

Target Loss 4 :

C3H4O2

Carbon Range 1 From : C30 to C27

Carbon Range 2 From : C20 to C17 ✕

Target Loss 5 :

C1H0O2

Carbon Range 1 From : C30 to C29

Carbon Range 2 From : C29 to C28 ✕

Loss Name 6: C2 loss

Target Loss 6 :

C2H4O2

Carbon Range 1 From : C32 to C30

Carbon Range 2 From : C22 to C20 ✕

Loss Name 7: C7 loss

Target Loss 7 :

C7H6O2

Carbon Range 1 From : C37 to C30

Carbon Range 2 From : C27 to C20 ✕

(4) Generate Extraction: Click **"Run Neutral Loss Extraction"** to label target neutral losses within certain carbon range, and the python console would be displayed the following information:

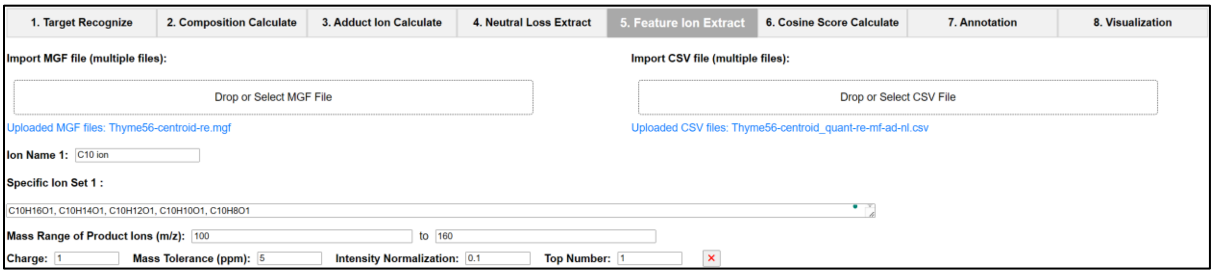
```
Run    CNPs-MFSA_Web ×

Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 18, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 16, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 14, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 12, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 10, 'O': 2}
Difference between C27H6008 and C17H1600: {'C': 10, 'H': 44, 'O': 8}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 18, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 16, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 14, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 12, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 10, 'O': 2}
```

(5) Export Outputs: download the generated **xxx.zip** file and the results were shown below with new rows, such as **C10 loss**, **C14 loss**, and so on. The **xxx-nl.csv** was used as the input file and switched to the next tab "Feature Ion Extract".

- Thyme56-centroid.mgf
- Thyme56-centroid-re.mgf
- Thyme56-centroid_quant.csv
- Thyme56-centroid_quant-re.csv
- Thyme56-centroid_quant-re-mf.csv
- Thyme56-centroid_quant-re-mf-ad.csv
- Thyme56-centroid_quant-re-mf-ad-nl.csv

	A	B	C	D	E	F	G	H	I	J	K	BP	BQ	BR	BS	BT	BU	BV
1	row ID	row m/z	row retent	D Score	MD Score	type	same peal	elemental	adduct ion	identificat	same com	C10 loss	C14 loss	C9 loss	C3 loss	CO2 loss	C2 loss	C7 loss
2	1418	680.2911	1.1	8	0	D	none	C33H45O1	[M+NH4]-	possibly u	none	none	none	none	C3H4O2	none	none	C7H6O2
3	444	512.2862	1.2	5	0	D	none	C26H41O9	[M+NH4]-	possibly k	daphnedit	none	none	none	C3H4O2	none	none	none
4	9757	682.4177	1.2	0	4	MD	none	C36H59O1	[M+NH4]-	possibly u	none	none	none	none	none	none	none	none
5	3143	682.4152	1.3	0	5	MD	none	C36H59O1	[M+NH4]-	possibly u	none	none	none	none	none	none	none	none
6	10205	469.2223	1.5	6	0	D	10193, 10:	C27H32O7	[M+H-H2(	possibly u	none	none	none	none	C3H4O2	none	none	C7H6O2
7	10193	487.2324	1.5	7	0	D	10193, 10:	C27H34O8	[M+H]+	possibly k	orthobenz	none	none	none	C3H4O2	none	none	C7H6O2
8	1389	680.2908	1.6	13	0	D	none	C33H45O1	[M+NH4]-	possibly u	none	none	none	none	C3H4O2	none	none	none
9	8407	329.1743	1.7	6	0	D	none	C20H24O4	[M+H]+	possibly u	none	none	none	none	none	none	none	none
10	7378	374.22	1.7	5	0	D	none	C19H35O6	[M+NH4]-	possibly u	none	none	none	none	C3H4O2	none	none	none
11	10189	451.2111	1.7	6	0	D	none	C27H30O6	[M+H]+	possibly u	none	none	none	none	C3H4O2	none	none	C7H6O2
12	10183	522.2697	1.7	6	0	D	none	C27H39O9	[M+NH4]-	possibly k	neogenkw	none	none	none	C3H4O2	none	none	C7H6O2
13	3182	359.1484	1.9	5	0	D	none	C20H22O6	[M+H]+	possibly u	none	none	none	none	none	none	none	none
14	8456	505.2432	2	11	0	D	none	C27H36O9	[M+H]+	possibly k	neogenkw	none	none	none	C3H4O2	none	none	none
15	10405	521.2385	2	7	0	D	none	C27H36O1	[M+H]+	possibly u	none	none	none	none	C3H4O2	none	none	none

Step 5: Feature Ion Labeling



(1) Upload MGF Files: Select **xxx-re.mgf** files processed by **Step 1**.

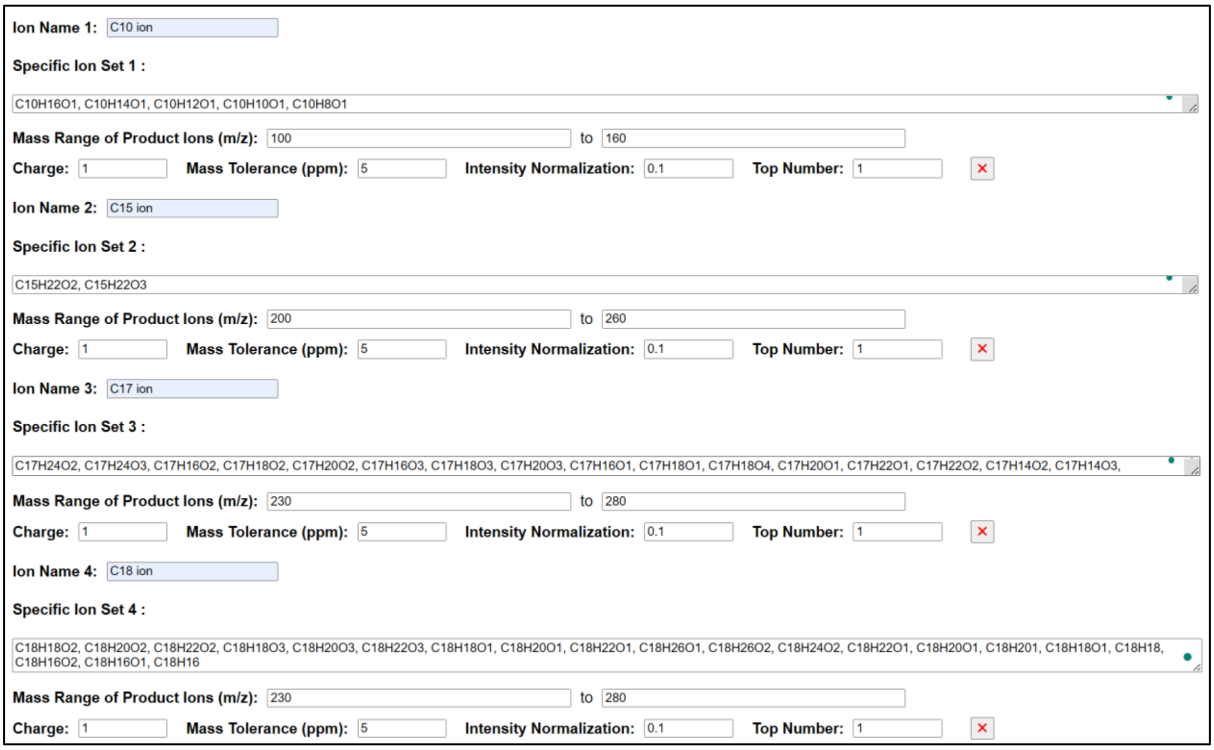
(2) Upload CSV Files: Select **xxx-nl.csv** files processed by **Step 4**.

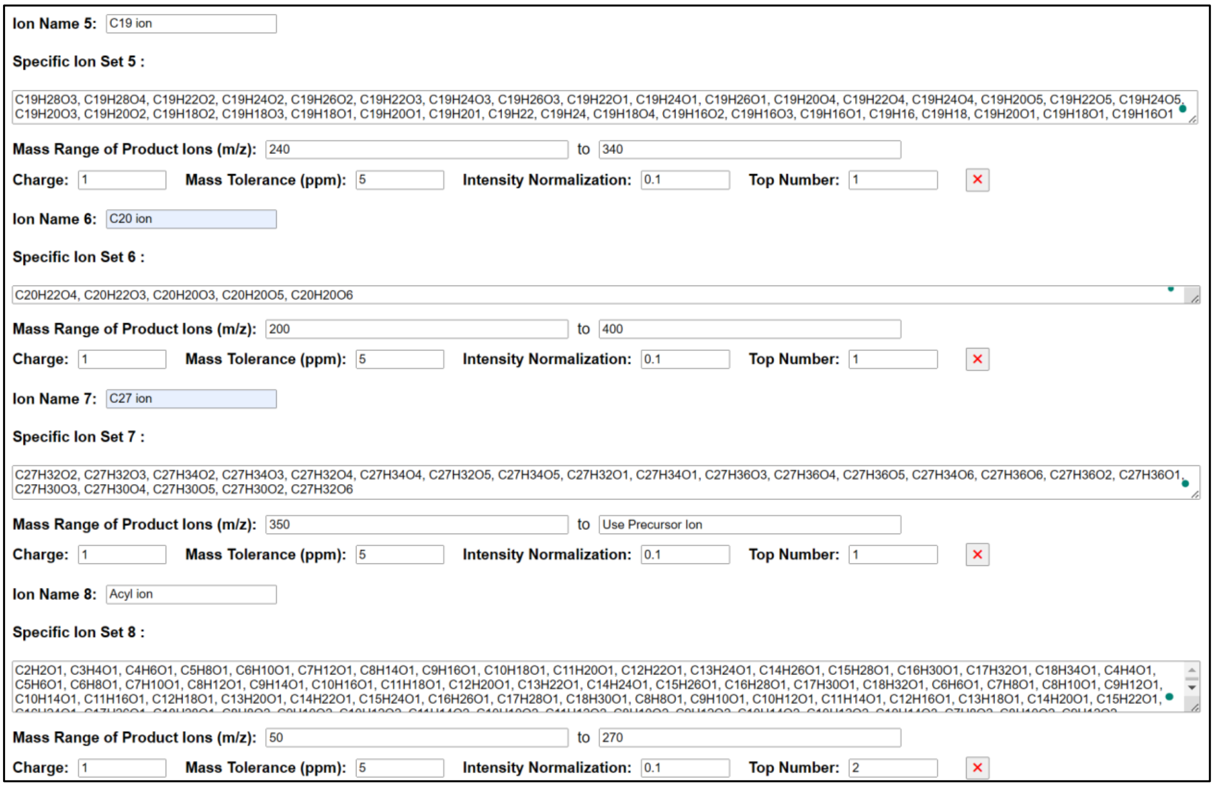
(3) Define Parameters: Input the diagnostic ion or ion set, and set up detailed **Mass range**, **Charge**, **Mass Tolerance (ppm)**, **Intensity Normalization**, and **Top Number** as following:

***Notes**:

- **Intensity Normalization > 0.1** was used to exclude product ions with relative intensities exceeding 10% within a certain mass range.

- **Top Number N** (n=1-10) represented the top N product ions of extraction intensity from highest to lowest. By modifying **Top Number N**, candidate diagnostic ion can be obtained according to the intensity of the ions, which could use for the generation of candidate structure in the subsequent "annotation" tab.

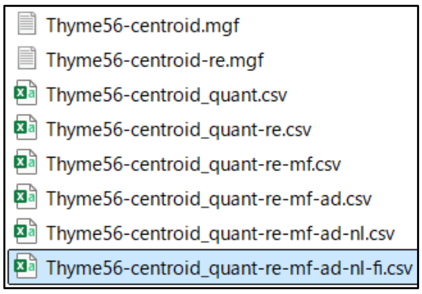


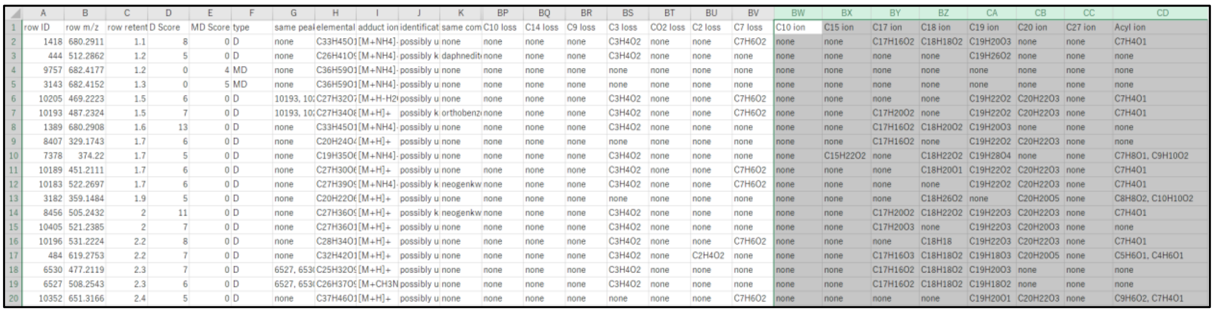


***Note**: For detailed ion information, see the **feature ion and neutral loss.txt** file, which can be find in GitHub repository CNPs-MFSA.

(4) Generate Extraction: Click **"Run Feature Ion Extraction"** to label target feature ion within certain carbon range.

(5) Export Outputs: download the generated **xxx.zip** file and the results were shown below with new rows, such as **C10 ion**, **C15 ion**, and so on. The **xxx-fi.csv** was used as the input file and switched to the next tab "Cosine Score Calculation".

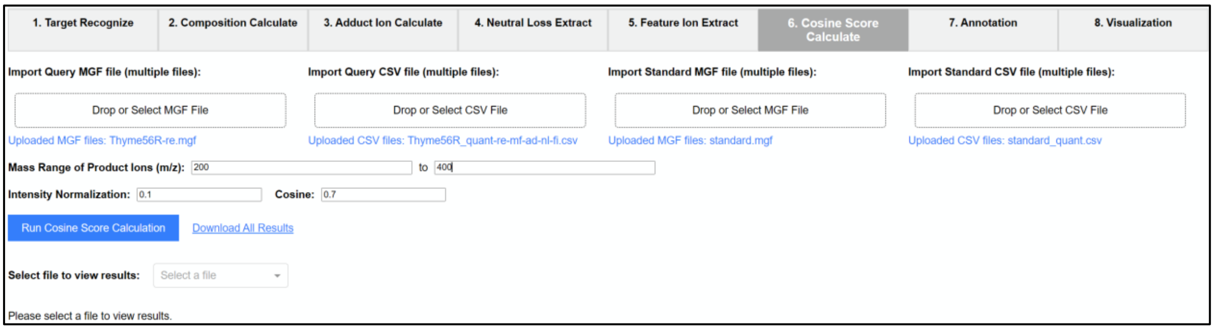




	A	B	C	D	E	F	G	H	I	J	K
1	row ID	row m/z	row retent	D Score	MD Score	type	same peal	elemental	adduct ion	identificat	same com
2	1418	680.2911	1.1	8	0	D	none	C33H45O1	[M+NH4]	possibly u	none
3	444	512.2862	1.2	5	0	D	none	C26H41O9	[M+NH4]	possibly k	daphnedit
4	9757	682.4177	1.2	0	4	MD	none	C36H59O1	[M+NH4]	possibly u	none
5	3143	682.4152	1.3	0	5	MD	none	C36H59O1	[M+NH4]	possibly u	none
6	10205	469.2223	1.5	6	0	D	10193, 10	C27H32O7	[M+H-H2	possibly u	none
7	10193	487.2324	1.5	7	0	D	10193, 10	C27H34O8	[M+H]+	possibly k	orthobenz
8	1389	680.2908	1.6	13	0	D	none	C33H45O1	[M+NH4]	possibly u	none
9	8407	329.1743	1.7	6	0	D	none	C20H24O4	[M+H]+	possibly u	none
10	7378	374.22	1.7	5	0	D	none	C19H35O6	[M+NH4]	possibly u	none
11	10189	451.2111	1.7	6	0	D	none	C27H30O6	[M+H]+	possibly u	none
12	10183	522.2697	1.7	6	0	D	none	C27H39O9	[M+NH4]	possibly k	neogenkw
13	3182	359.1484	1.9	5	0	D	none	C20H22O6	[M+H]+	possibly u	none
14	8456	505.2432	2	11	0	D	none	C27H36O9	[M+H]+	possibly k	neogenkw
15	10405	521.2385	2	7	0	D	none	C27H36O1	[M+H]+	possibly u	none
16	10196	531.2224	2.2	8	0	D	none	C28H34O1	[M+H]+	possibly u	none
17	484	619.2753	2.2	7	0	D	none	C32H42O1	[M+H]+	possibly u	none
18	6530	477.2119	2.3	7	0	D	6527, 653	C25H32O9	[M+H]+	possibly u	none
19	6527	508.2543	2.3	6	0	D	6527, 653	C26H37O9	[M+CH3N	possibly u	none
20	10352	651.3166	2.4	5	0	D	none	C37H46O1	[M+H]+	possibly u	none

	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD
1	C10 loss	C14 loss	C9 loss	C3 loss	CO2 loss	C2 loss	C7 loss	C10 ion	C15 ion	C17 ion	C18 ion	C19 ion	C20 ion	C27 ion	Acyl ion
2	none	none	none	C3H4O2	none	none	C7H6O2	none	none	C17H16O2	C18H18O2	C19H20O3	none	none	C7H4O1
3	none	none	none	C3H4O2	none	none	none	none	none	none	none	C19H26O2	none	none	none
4	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none
5	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none
6	none	none	none	C3H4O2	none	none	C7H6O2	none	none	none	none	C19H22O2	C20H22O3	none	C7H4O1
7	none	none	none	C3H4O2	none	none	C7H6O2	none	none	C17H20O2	none	C19H22O2	C20H22O3	none	C7H4O1
8	none	none	none	C3H4O2	none	none	none	none	none	C17H16O2	C18H20O2	C19H20O3	none	none	none
9	none	none	none	none	none	none	none	none	none	C17H16O2	none	C19H22O2	C20H22O3	none	none
10	none	none	none	C3H4O2	none	none	none	none	C15H22O2	none	C18H22O2	C19H28O4	none	none	C7H8O1, C9H10O2
11	none	none	none	C3H4O2	none	none	C7H6O2	none	none	none	C18H20O1	C19H22O2	C20H22O3	none	C7H4O1
12	none	none	none	C3H4O2	none	none	C7H6O2	none	none	none	none	C19H22O2	C20H22O3	none	C7H4O1
13	none	none	none	none	none	none	none	none	none	none	C18H26O2	none	C20H20O5	none	C8H8O2, C10H10O2
14	none	none	none	C3H4O2	none	none	none	none	none	C17H20O2	C18H22O2	C19H22O3	C20H22O3	none	C7H4O1
15	none	none	none	C3H4O2	none	none	none	none	none	C17H20O3	none	C19H22O3	C20H20O3	none	none
16	none	none	none	C3H4O2	none	none	C7H6O2	none	none	none	C18H18	C19H22O3	C20H22O3	none	C7H4O1
17	none	none	none	C3H4O2	none	C2H4O2	none	none	none	C17H16O3	C18H18O2	C19H18O3	C20H20O5	none	C5H6O1, C4H6O1
18	none	none	none	C3H4O2	none	none	none	none	none	C17H16O2	C18H18O2	C19H20O3	none	none	none
19	none	none	none	C3H4O2	none	none	none	none	none	C17H16O2	C18H18O2	C19H18O2	none	none	none
20	none	none	none	none	none	none	C7H6O2	none	none	none	none	C19H20O1	C20H22O3	none	C9H6O2, C7H4O1

Step 6: Cosine Score Calculate



(1) Upload Query MGF Files: Select **xxx-re.mgf** files processed by **Step 1**.

(2) Upload Query CSV Files: Select **xxx-nl.csv** files processed by **Step 5**.

(3) Upload Standard MGF Files: Select **standard.mgf** files.

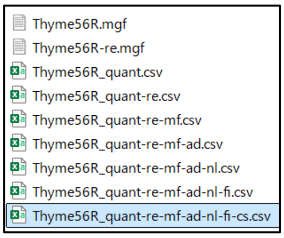
(4) Upload Standard CSV Files: Select **standard_quant.csv** files.

***Notes:** The input standard CSV files should be prepared as the following format with **standards name** and corresponding **substructure type**.

	A	B	C	D	E	F
1	row ID	row m/z	row retention time	peak area	standards name	substructure type
2	1	591.2586	8.3951723	3.95E+09	genkwanine M	6,7-epoxy
3	2	591.2588	8.7453353	2.59E+08	genkwanine N	6,7-epoxy
4	3	533.3109	11.595447	8.44E+08	simplexin	6,7-epoxy
5	4	609.2679	8.3171075	5.39E+09	genkwanine H	6,7-diol
6	5	609.2679	8.0269279	1.30E+07	genkwanine G	6,7-diol
7	6	573.2477	4.1568114	2.67E+08	DPSft27	4,6-epoxy
8	7	649.3006	10.827824	1.59E+09	yuanhuacine	6,7-epoxy
9	8	609.269	3.8598187	7.17E+07	Wlichi7 (3.86)	4,7-epoxy
10	9	609.269	6.2090547	8.10E+07	Wlichi7 (6.20)	4,7-epoxy

(5) Generate Cosine Score: Click **"Run Cosine Score Calculation"** to calculate.

(6) Export Outputs: download the generated **xxx.zip** file and the results were shown below with new rows, **<u>cosine score</u>** and **<u>substructure type</u>** (or modified as **B part** for daphnanes). The **xxx-cs.csv** was used as the input file and switched to the next tab "Annotation".



	A	B	C	D	E	F	G	H	I	J	K
1	row ID	row m/z	row retent	D Score	MD Score	type	same peal	elemental	adduct ion	identificat	same com
2	1418	680.2911	1.1	8	0	D	none	C33H45O1	[M+NH4]-	possibly u	none
3	444	512.2862	1.2	5	0	D	none	C26H41O	[M+NH4]-	possibly k	daphnedit
4	9757	682.4177	1.2	0	4	MD	none	C36H59O1	[M+NH4]-	possibly u	none
5	3143	682.4152	1.3	0	5	MD	none	C36H59O1	[M+NH4]-	possibly u	none
6	10205	469.2223	1.5	6	0	D	10193, 10:	C27H32O7	[M+H-H2	possibly u	none
7	10193	487.2324	1.5	7	0	D	10193, 10:	C27H34O	[M+H]+	possibly k	orthobenz
8	1389	680.2908	1.6	13	0	D	none	C33H45O1	[M+NH4]-	possibly u	none
9	8407	329.1743	1.7	6	0	D	none	C20H24O	[M+H]+	possibly u	none
10	7378	374.22	1.7	5	0	D	none	C19H35O	[M+NH4]-	possibly u	none
11	10189	451.2111	1.7	6	0	D	none	C27H30O	[M+H]+	possibly u	none
12	10183	522.2697	1.7	6	0	D	none	C27H39O	[M+NH4]-	possibly k	neogenkw
13	3182	359.1484	1.9	5	0	D	none	C20H22O	[M+H]+	possibly u	none
14	8456	505.2432	2	11	0	D	none	C27H36O	[M+H]+	possibly k	neogenkw
15	10405	521.2385	2	7	0	D	none	C27H36O1	[M+H]+	possibly u	none
16	10196	531.2224	2.2	8	0	D	none	C28H34O1	[M+H]+	possibly u	none
17	484	619.2753	2.2	7	0	D	none	C32H42O1	[M+H]+	possibly u	none
18	6530	477.2119	2.3	7	0	D	6527, 653	C25H32O	[M+H]+	possibly u	none
19	6527	508.2543	2.3	6	0	D	6527, 653	C26H37O	[M+CH3N	possibly u	none
20	10352	651.3166	2.4	5	0	D	none	C37H46O1	[M+H]+	possibly u	none

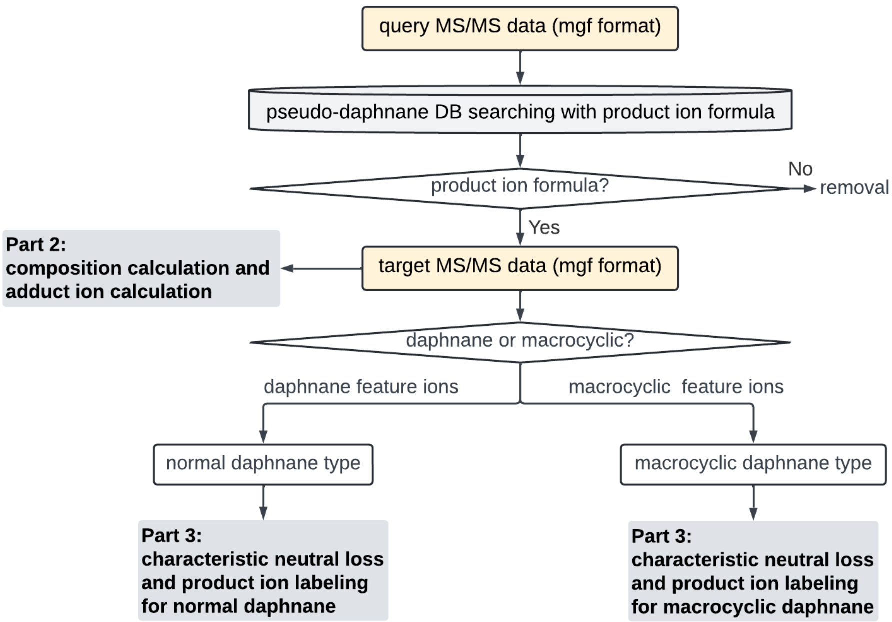
	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY
1	C10 loss	C14 loss	C9 loss	C3 loss	CO2 loss	C2 loss	C7 loss	C10 ion	C15 ion	C17 ion
2	none	none	none	C3H4O2	none	none	C7H6O2	none	none	C17H16O
3	none	none	none	C3H4O2	none	none	none	none	none	none
4	none	none	none	none	none	none	none	none	none	none
5	none	none	none	none	none	none	none	none	none	none
6	none	none	none	C3H4O2	none	none	C7H6O2	none	none	none
7	none	none	none	C3H4O2	none	none	C7H6O2	none	none	C17H20O
8	none	none	none	C3H4O2	none	none	none	none	none	C17H16O
9	none	none	none	none	none	none	none	none	none	C17H16O
10	none	none	none	C3H4O2	none	none	none	none	C15H22O	none
11	none	none	none	C3H4O2	none	none	C7H6O2	none	none	none
12	none	none	none	C3H4O2	none	none	C7H6O2	none	none	none
13	none	none	none	none	none	none	none	none	none	none
14	none	none	none	C3H4O2	none	none	none	none	none	C17H20O
15	none	none	none	C3H4O2	none	none	none	none	none	C17H20O
16	none	none	none	C3H4O2	none	none	C7H6O2	none	none	none
17	none	none	none	C3H4O2	none	C2H4O2	none	none	none	C17H16O
18	none	none	none	C3H4O2	none	none	none	none	none	C17H16O
19	none	none	none	C3H4O2	none	none	none	none	none	C17H16O
20	none	none	none	none	none	none	C7H6O2	none	none	none

	BZ	CA	CB	CC	CD	CE	CF
1	C18 ion	C19 ion	C20 ion	C27 ion	Acyl ion	cosine score	substructure type
2	C18H18O	C19H20O	none	none	C7H4O1	0.74	6,7-epoxy
3	none	C19H26O	none	none	none	value < 0.7	others
4	none	none	none	none	none	value < 0.7	others
5	none	none	none	none	none	value < 0.7	others
6	none	C19H22O	C20H22O	none	C7H4O1	0.83	4,7-epoxy
7	none	C19H22O	C20H22O	none	C7H4O1	0.72	4,7-epoxy
8	C18H20O	C19H20O	none	none	none	value < 0.7	others
9	none	C19H22O	C20H22O	none	none	value < 0.7	others
10	C18H22O	C19H28O	none	none	C7H8O1, (	value < 0.7	others
11	C18H20O1	C19H22O	C20H22O	none	C7H4O1	value < 0.7	others
12	none	C19H22O	C20H22O	none	C7H4O1	value < 0.7	others
13	C18H26O	none	C20H20O	none	C8H8O2, (	value < 0.7	others
14	C18H22O	C19H22O	C20H22O	none	C7H4O1	value < 0.7	others
15	none	C19H22O	C20H20O	none	none	value < 0.7	others
16	C18H18	C19H22O	C20H22O	none	C7H4O1	value < 0.7	others
17	C18H18O	C19H18O	C20H20O	none	C5H6O1, (	value < 0.7	others
18	C18H18O	C19H20O	none	none	none	0.79	6,7-epoxy
19	C18H18O	C19H18O	none	none	none	0.89	6,7-epoxy
20	none	C19H20O1	C20H22O	none	C9H6O2, (	0.71	4,7-epoxy

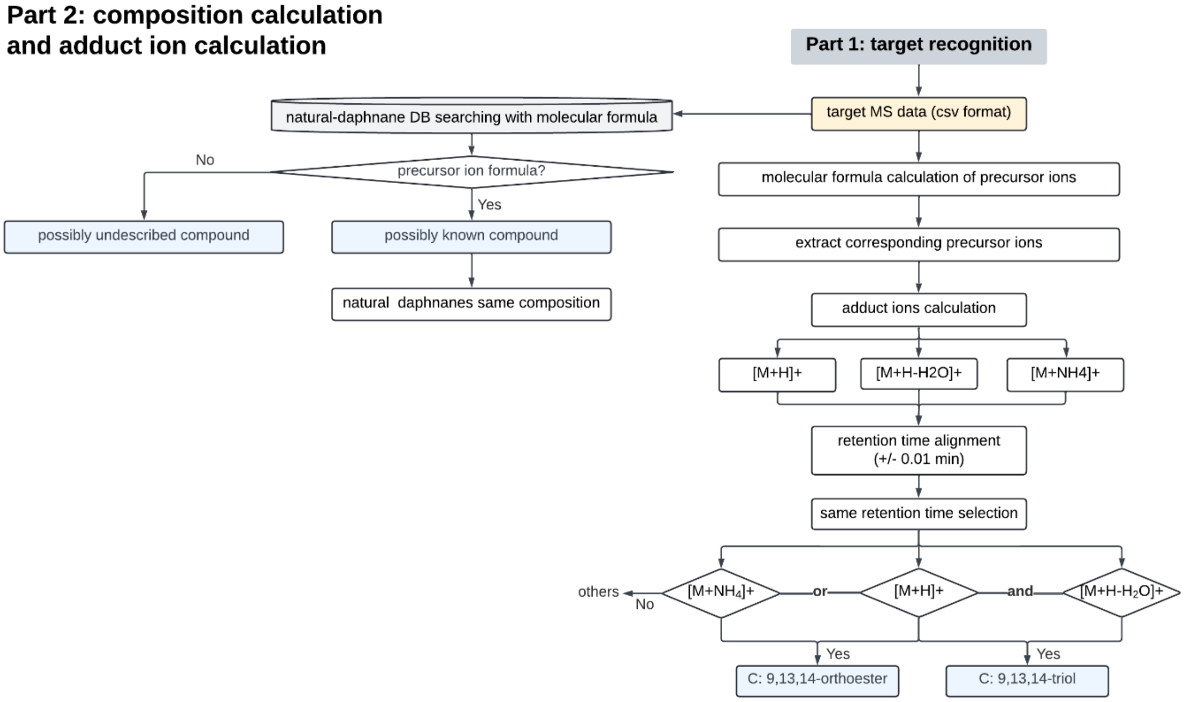
Step 7: Annotation for Daphnane

***Notes**: The current workflow was based on the following flowcharts. User could modify the source code of "Annotation" tab for module combination of other types of compounds.

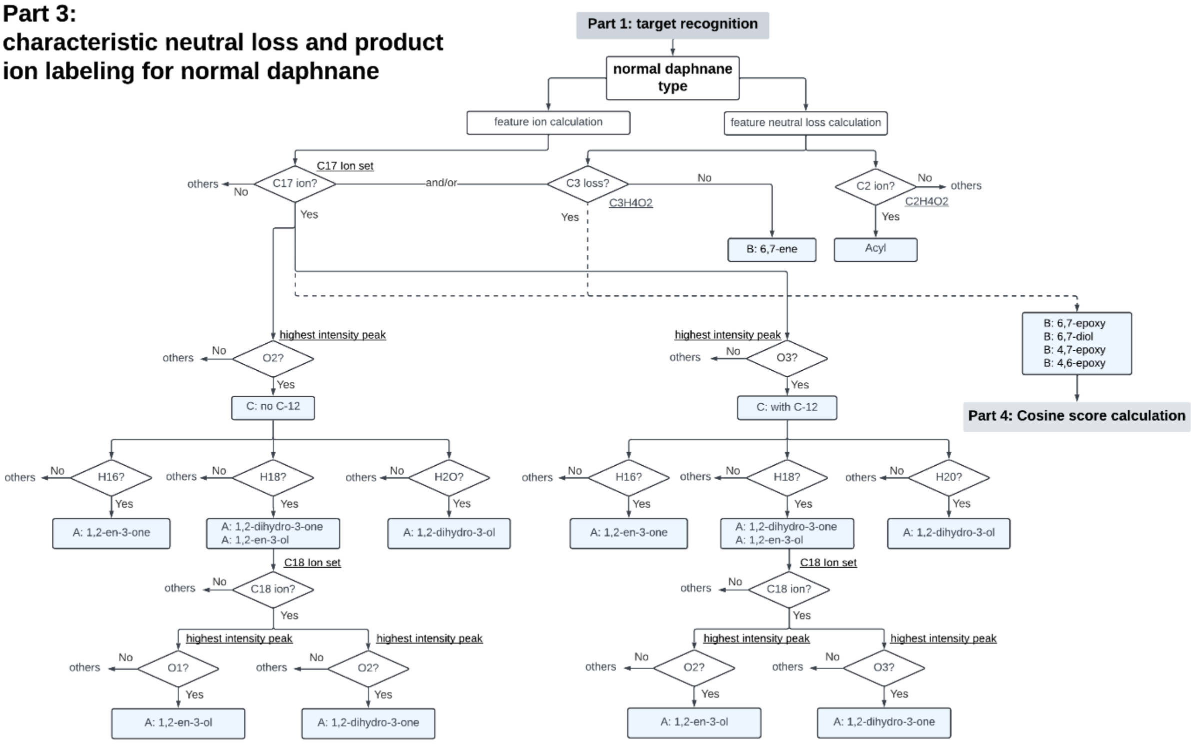
Part 1: target recognition



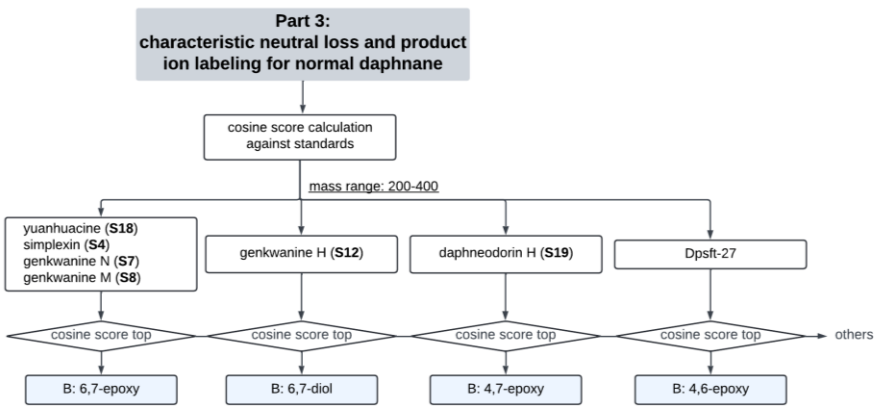
Part 2: composition calculation and adduct ion calculation



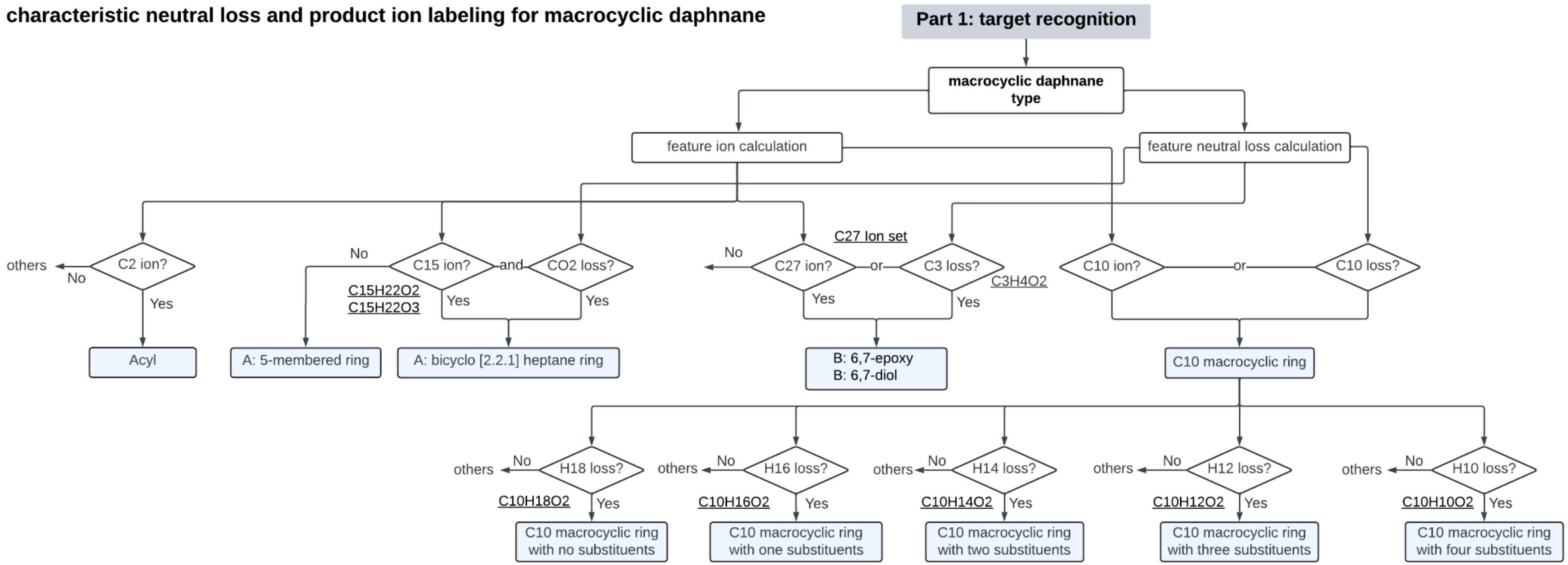
Part 3:
characteristic neutral loss and product ion labeling for normal daphnane



Part 4: Cosine score calculation within a certain mass range



Part 3:
characteristic neutral loss and product ion labeling for macrocyclic daphnane



(1) Upload CSV Files: Select **xxx-cs.csv** files processed by **Step 6**.

***Notes**: The column name **substructure type** should be changed to **B part** for daphnane.

(2) Upload DB Files: Select **Skeleton_DB.csv** and **Acyl_DB.csv** files generated by **"pesudo library construction"** files, which could be found in Github. The "pesudo library construction" files contained three parts and can used to generate **Skeleton_DB.csv** and **Acyl_DB.csv**, respectively.

- Acyl Structure Simulation and Feature Prediction
- Skeleton Feature Prediction
- Skeleton Structure Simulation

a). Skeleton_DB.csv

	A	B	C	D	E	F
1	Number	A parts	B parts	C parts	M parts	New SMILES
2	1	A1	B1	C1		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)CC(C)C1(O4)C1C=C(C)C(=O)C1(O)C2O
3	2	A2	B1	C1		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)CC(C)C1(O4)C1CC(C)C(=O)C1(O)C2O
4	3	A3	B1	C1		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)CC(C)C1(O4)C1C=C(C)C(O*)C1(O)C2O
5	4	A4	B1	C1		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)CC(C)C1(O4)C1CC(C)C(O*)C1(O)C2O
6	5	A5	B1	C1		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)CC(C)C1(O4)C1=CC(C)C(=O)C1(O)C2O
7	6	A1	B1	C2		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1C=C(C)C(=O)C1(O)C2O
8	7	A2	B1	C2		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1CC(C)C(=O)C1(O)C2O
9	8	A3	B1	C2		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1C=C(C)C(O*)C1(O)C2O
10	9	A4	B1	C2		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1CC(C)C(O*)C1(O)C2O
11	10	A5	B1	C2		*OCC12OC1C1C3OC4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1=CC(C)C(=O)C1(O)C2O
12	11	A1	B1	C3		*OCC12OC1C1C(O*)C(O)(C(=C)C)CC(C)C1(O)C1C=C(C)C(=O)C1(O)C2O
13	12	A2	B1	C3		*OCC12OC1C1C(O*)C(O)(C(=C)C)CC(C)C1(O)C1CC(C)C(=O)C1(O)C2O
14	13	A3	B1	C3		*OCC12OC1C1C(O*)C(O)(C(=C)C)CC(C)C1(O)C1C=C(C)C(O*)C1(O)C2O
15	14	A4	B1	C3		*OCC12OC1C1C(O*)C(O)(C(=C)C)CC(C)C1(O)C1CC(C)C(O*)C1(O)C2O

b). Acyl_DB.csv

	A	B	C	D	E
1	no.	R parts	Acyl SMILES	Acyl Formula	Identification
2	1	R1	CC(=O)*	C2H2O1	acyl
3	2	R2	CCC(=O)*	C3H4O1	propionoyl
4	3	R3	CCCC(=O)*	C4H6O1	butanoyl
5	4	R4	CCCCC(=O)*	C5H8O1	pentanoyl
6	5	R5	CCCCCC(=O)*	C6H10O1	hexanoyl
7	6	R6	CCCCCCC(=O)*	C7H12O1	heptanoyl
8	7	R7	CCCCCCCC(=O)*	C8H14O1	octanoyl
9	8	R8	CCCCCCCCC(=O)*	C9H16O1	nonanoyl
10	9	R9	CCCCCCCCCC(=O)*	C10H18O1	decanoyl

(3) Generate Annotation: Click **"Run Annotation"** to calculate.

(4) Export Outputs: download the generated **xxx.zip** file and the results were shown below with new rows, such as the **substructure information** and **the combination skeleton and substituents information**. The **xxx-an (index).csv** was used as the input file and switched to the next tab "Visualization".

- Thyme56-centroid.mgf
- Thyme56-centroid-re.mgf
- Thyme56-centroid_quant.csv
- Thyme56-centroid_quant-re.csv
- Thyme56-centroid_quant-re-mf.csv
- Thyme56-centroid_quant-re-mf-ad.csv
- Thyme56-centroid_quant-re-mf-ad-nl.csv
- Thyme56-centroid_quant-re-mf-ad-nl-fi.csv
- Thyme56-centroid_quant-re-mf-ad-nl-fi-cs.csv
- Thyme56-centroid_quant-re-mf-ad-nl-fi-cs-an (index).csv
- Thyme56-centroid_quant-re-mf-ad-nl-fi-cs-an.csv

substructure information **the combination information**

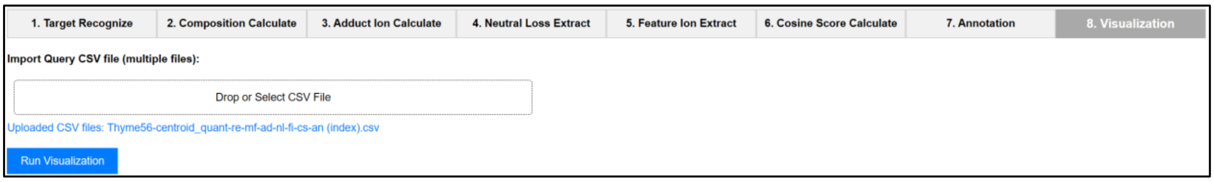
	CF	CG	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR
1	A part	B part	C part	C-12	M part	Acyl part	A number	B number	C number	M number	Acyl Number	Acyl SMILES	Skeleton SMILES
2	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	5-pheny-2,4-pentad	A1	others	C2	others	R83,R79	c1ccccc1C=CC=CC	others
3	1,2-dihydro-3-ol	4,7-epoxy	9,13,14-orthoester	no C-12	others	benzoyl	A4	B7	C1	others	R79	c1ccccc1C(=O)*	others
4	1,2-dihydro-3-one	others	9,13,14-orthoester	with C-12	others	5-pheny-2,4-pentad	A2	others	C2	others	R83,R79	c1ccccc1C=CC=CC	others
5	1,2-en-3-one	6,7-epoxy	9,13,14-orthoester	with C-12	others	decadienoyl,benzoy	A1	B1	C2	others	R37,R79	CCCCCC=CC=CC(=	*OCC12OC1C1C3O
6	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	cinnamoyl,benzoyl	A1	others	C2	others	R80,R79	c1ccccc1C=CC(=O)	others
7	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	decatrienoyl,benzoy	A1	others	C2	others	R48,R79	CCCC=CC=CC=CC	others
8	1,2-dihydro-3-ol	others	9,13,14-orthoester	no C-12	others	5-pheny-2,4-pentad	A4	others	C1	others	R83	c1ccccc1C=CC=CC	others
9	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	benzoyl	A1	others	C2	others	R79	c1ccccc1C(=O)*	others
10	1,2-dihydro-3-ol	6,7-epoxy	9,13,14-orthoester	no C-12	others	decadienoyl	A4	B1	C1	others	R37	CCCCCC=CC=CC(=	*OCC12OC1C1C3O
11	1,2-dihydro-3-ol	6,7-diol	9,13,14-orthoester	no C-12	others	benzoyl	A4	B2	C1	others	R79	c1ccccc1C(=O)*	*OCC1(O)C(O)C2C3
12	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	5-pheny-2,4-pentad	A1	others	C2	others	R83,R79	c1ccccc1C=CC=CC	others
13	others	others	9,13,14-orthoester	others	others	benzoyl	others	others	others	others	R79	c1ccccc1C(=O)*	others
14	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	5-pheny-2,4-pentad	A1	others	C2	others	R83,R79	c1ccccc1C=CC=CC	others
15	others	others	9,13,14-orthoester	others	others	acyl,benzoyl	others	others	others	others	R1,R79	CC(=O)*,c1ccccc1C	others
16	others	others	9,13,14-triol	others	others	5-pheny-2,4-pentad	others	others	others	others	R83,R79	c1ccccc1C=CC=CC	others
17	1,2-dihydro-3-one	others	9,13,14-orthoester	with C-12	others	benzoyl	A2	others	C2	others	R79	c1ccccc1C(=O)*	others
18	1-alkyl-3-ol	6,7-epoxy	9,13,14-orthoester	others	C10 ring with four s	benzoyl	A8	B1	C1	M6	R79	c1ccccc1C(=O)*	*OCC12OC1C1C3O
19	1-alkyl-3-ol	6,7-epoxy	9,13,14-orthoester	others	C10 ring with one su	benzoyl	A8	B1	C1	M3	R79	c1ccccc1C(=O)*	*OCC12OC1C1C3O
20	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	5-pheny-2,4-pentad	A1	others	C2	others	R83,R79	c1ccccc1C=CC=CC	others
21	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	benzoyl	A1	others	C2	others	R79	c1ccccc1C(=O)*	others
22	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	acyl,5-pheny-2,4-pe	A1	others	C2	others	R1,R83	CC(=O)*,c1ccccc1C	others
23	1,2-dihydro-3-one	others	9,13,14-orthoester	with C-12	others	decatrienoyl,benzoy	A2	others	C2	others	R48,R79	CCCC=CC=CC=CC	others
24	1,2-dihydro-3-ol	4,6-epoxy	9,13,14-orthoester	no C-12	others	benzoyl	A4	B8	C1	others	R79	c1ccccc1C(=O)*	others
25	1-alkyl-3-ol	6,7-epoxy	9,13,14-orthoester	others	C10 ring with three	acyl,benzoyl,nonenc	A8	B1	C1	M5	R1,R79,R23	CC(=O)*,c1ccccc1C	*OCC12OC1C1C3O
26	others	others	9,13,14-orthoester	others	others	benzoyl	others	others	others	others	R79	c1ccccc1C(=O)*	others
27	1,2-en-3-one	others	9,13,14-orthoester	no C-12	others	decadienoyl,hexadie	A1	others	C1	others	R37,R33	CCCCCC=CC=CC(=	others
28	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	cinnamoyl,benzoyl	A1	others	C2	others	R80,R79	c1ccccc1C=CC(=O)	others
29	1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	benzoyl	A1	others	C2	others	R79	c1ccccc1C(=O)*	others
30	others	others	9,13,14-orthoester	others	others	acyl,benzoyl,decadi	others	others	others	others	R1,R79,R37	CC(=O)*,c1ccccc1C	others

Step 8: Visualization

(1) Upload CSV Files: Select **xxx-an (index).csv** files processed by **Step 7**.

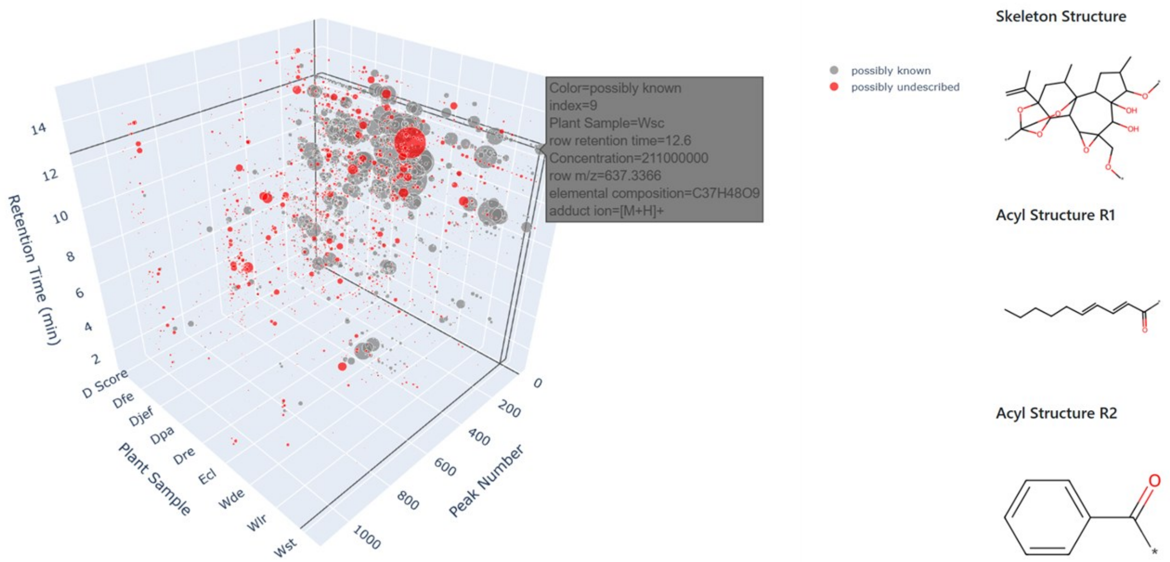
***Notes**: Since the **"row ID"** in the files output in the previous steps were discontinuous serial numbers, an **"index"** column should be inserted **manually** before conducting the last step.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	index	row ID	row m/z	row retent	20221105_	20221105_	20221207_	20221105_	20221207_	20221105_	20221105_	20221207_	20221105_	20221207_	20221207_	20230217_
2	1	1	655.2525	8.7	8.47E+08	1.09E+08	0	0	0	0	0	0	0	0	0	0
3	2	2	591.2585	8.8	5.76E+08	2.44E+09	25100000	68200000	17700000	254145.6	4709378	14700000	2251164	5619186	6663405	26000000
4	3	3	657.2678	8.9	97400000	1.18E+08	0	0	0	0	0	0	0	0	0	0
5	4	5	649.3003	10.7	54400000	3810075	1.04E+08	15400000	12200000	555498.9	16000000	1537602	794179.9	2715672	0	0
6	5	6	629.2382	7.7	58400000	0	0	1.19E+08	70300000	6504685	4538744	4795256	0	0	0	0
7	6	7	647.2847	10	62600000	9713001	18400000	1.31E+09	6.43E+08	1.68E+08	9.33E+08	89600000	0	4.45E+08	0	0
8	7	8	643.2881	10.8	1.13E+08	1.71E+09	676779	396669.9	0	0	488109.4	188933.6	0	0	0	0
9	8	9	603.2223	6.8	44100000	0	60500000	2380547	70200000	0	4175097	2622250	0	0	0	0
10	9	10	637.3366	12.6	47400000	2.65E+08	5415838	0	7999909	0	2830324	1902679	0	2862820	0	7213202



(2) Generate Visualization: Click **"Run Visualization"** to calculate.

- **3D interactive scatter plot** for viewing the overall picture of the plant sample dataset

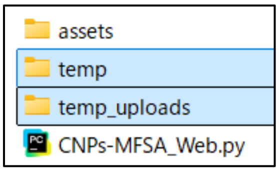


4. Tips for Effective Usage

(1) Download Files: If users have questions about downloading files from the web-application, the results files could be found in the following path:

C:/Users/**username**/PycharmProjects/CNPs-MFSA/temp or

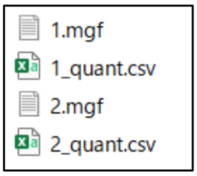
C:/Users/**username**/PycharmProjects/CNPs-MFSA/temp_uploads



(2) Input File Format: Ensure all input files are correctly formatted. Supported formats including xxx**.mgf** for spectral data, xxx**.csv** for metadata, and xxx**.db** for formula database.

(3) Error Handling: If a process fails, verify file integrity and parameter settings before re-running. Avoid repeated uploads. If it does not work, restart and rerun.

(4) For multiple files analysis, the file name of xxx**.mgf** and xxx**.csv** should be the same.



5. Support and Documentation

- GitHub Repository: Visit our repository [https://github.com/MiZhang47/CNPs-MFSA.git] for detailed documentation and issue tracking.

- Email Support: Any question or support about CNPs-MFSA could contact to Dr. Mi Zhang (zhangmi495@gmail.com).